



PROBABILITY THEORY

MTM2592-Gr:2

THIRD WEEK

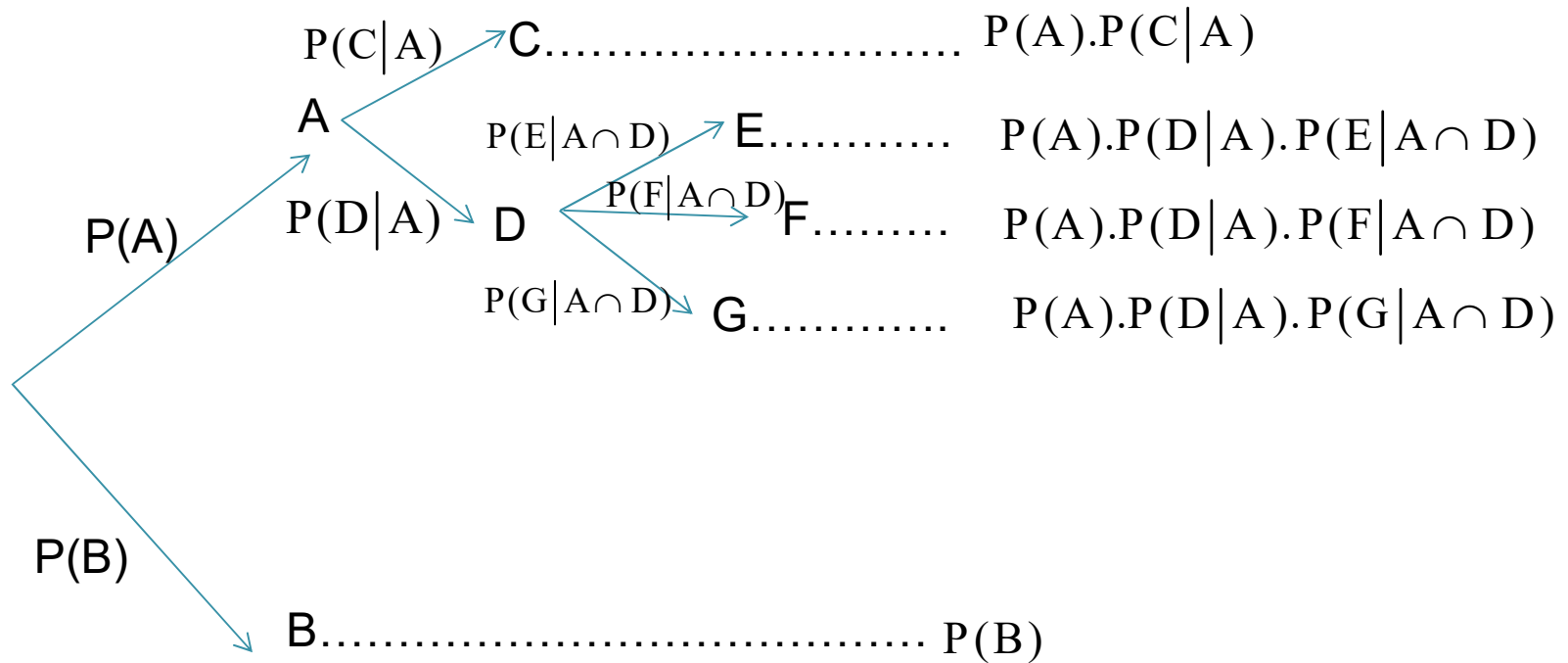
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DETERMINATION OF CONDITIONAL PROBABILITY BY TREE DIAGRAMS:





Example: There are 3 white balls and 5 red balls in a box. 2 balls are successively selected from the box, without replacing them back. Find the probability that two balls are different coloured by using tree diagram.

Solution:

Example: There are 4 black and 6 red balls in the first box and 3 black and 2 red balls in the second box. If we randomly select a box, then randomly select a ball from the chosen box, what is the probability that

- a) The chosen ball is black,
- b) The chosen box is the first one given that a black ball is selected?

Solution:

BAYES' THEOREM:

Let $B_1, B_2, \dots, B_n$ be discretization of a sample space S and A be an event in S with $P(A) \neq 0$

Then,

$$P(B_r | A) = \frac{P(B_r) \cdot P(A | B_r)}{\sum_{i=1}^n P(B_i) P(A | B_i)}$$

Proof: Note:

$$P(A) = \sum_{i=1}^n P(B_i \cap A) = \sum_{i=1}^n P(B_i) \cdot P(A | B_i)$$

$$P(B_r | A) = \frac{P(A \cap B_r)}{P(A)} = \frac{P(B_r) \cdot P(A | B_r)}{\sum_{i=1}^n P(B_i) P(A | B_i)}$$

Note: Bayes' Theorem is based on obtaining an unknown $P(B | A)$ conditional probability using the known $P(A | B)$ conditional probability.



Example: In a factory, three machines B_1 , B_2 , B_3 , make 30%, 45% and 25%, respectively of the products. It is known that 2%, 3% and 2% of the products made by each machine, respectively are defective.

- a) What is the probability that a randomly selected product is defective?
- b) If a randomly selected product is defective, what is the probability that it was made by machine B_3 ?

Solution:



Example: Members of a consulting firm rent cars from three companies, 60% from Company 1, 30% from Company 2, and 10% from Company 3. If 9% of vehicles from Company 1, 20% of vehicles from Company 2 and 6% of vehicles from Company 3 require maintenance,

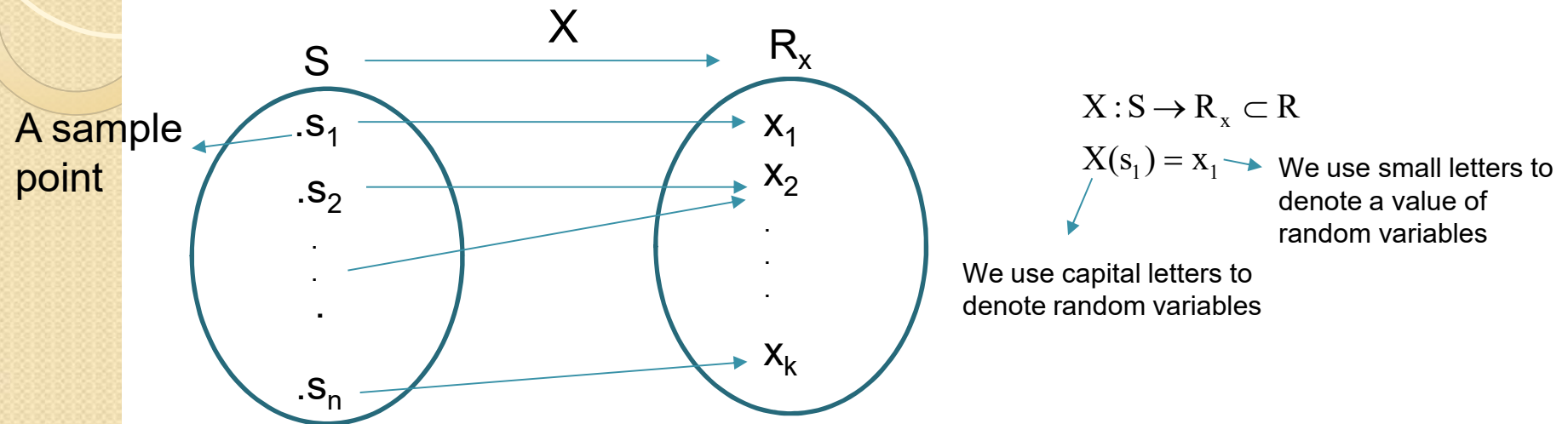
a) What is the probability that a car rented to the company will require maintenance?

b) If the car rented to the company requires maintenance, what is the probability that this vehicle came from Company 2?

Solution:

RANDOM VARIABLE

Definition: A random variable is a function from the sample space to the set of real numbers.



Ex: If a coin tossed until it shows head and the random variable is the number of trials, then,

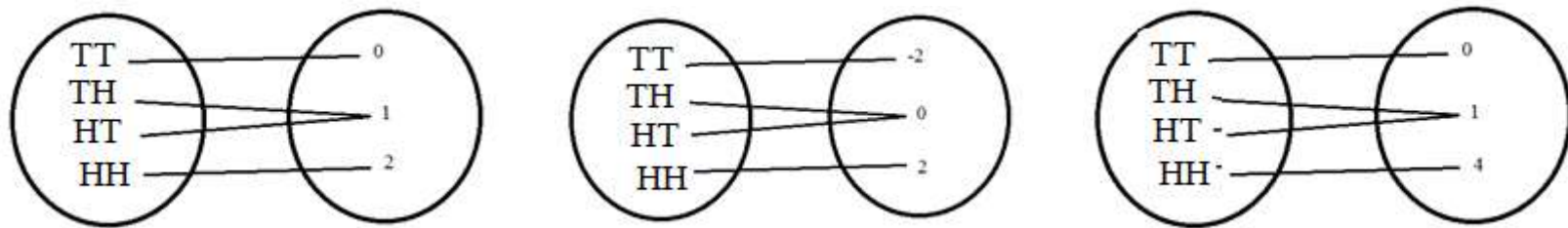
$$S = \{H, TH, TTH, TTTH, \dots\}$$

$$R_x = \{1, 2, 3, 4, \dots\}$$

Ex: A coin is tossed two times. Let random variable X shows the number of heads, random variable Y shows the difference of the number of tails from the number of heads, and random variable Z shows the square of the number of heads. Create definition and image sets of these random variables.

Definition Set $S = \{TT, TH, HT, HH\}$

Image Sets: $X(S) = \{0, 1, 2\}$, $Y(S) = \{-2, 0, 2\}$, $Z(S) = \{0, 1, 4\}$



Ex: Let two coins are tossed. Random variable Y shows the number of heads. There are 4 possible results

$$P\{Y = 0\} = P\{(T, T)\} = \frac{1}{4} \quad P\{Y = 0\} + P\{Y = 1\} + P\{Y = 2\} = 1 \text{ and } P(Y = y) \geq 0$$

$$P\{Y = 1\} = P\{(H, T), (T, H)\} = \frac{2}{4}$$

$$P\{Y = 2\} = P\{(H, H)\} = \frac{1}{4}$$

Ex: Rolling of a pair of dies.

- a) X: the sum of numbers
- b) X: the absolute value of the difference of numbers

$$S = \{(x_1, x_2) : x_i = 1, 2, 3, 4, 5, 6; i = 1, 2\}$$

$$\text{a) } R_x = \{X : x = x_1 + x_2\} = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$\text{b) } R_x = \{X : x = |x_1 - x_2|\} = \{0, 1, 2, 3, 4, 5\}$$

If R_x that contains value of the random variable X is finite or countably infinite, then X is said to be discrete. If R_x is an interval or a set of intervals, then X is said to be continuous.

Ex: Two socks were randomly selected from a drawer containing 5 brown and 3 green socks, successively. List the elements of sample space, the corresponding probabilities and the corresponding values w of the random variable W , where W the number of brown socks selected.

Solution:

Sample Space	Probability	w
BB	$5/8 \cdot 4/7 = 5/14$	2
BG	$5/8 \cdot 3/7 = 15/56$	1
GB	$3/8 \cdot 5/7 = 15/56$	1
GG	$3/8 \cdot 2/7 = 3/28$	0

Also, we can write $P(W = 2) = \frac{5}{14}$, for example, for the probability of the event that the random variable W will take on the value 2.

Ex: A balanced coin is tossed four times. List the elements of the sample space that are presumed to be equally likely, as this is what we mean by a coin being balanced, and the corresponding values x of the random variable X , the total number of heads.

Solution

If H and T stand for heads and tails, the results are as shown in the following table:

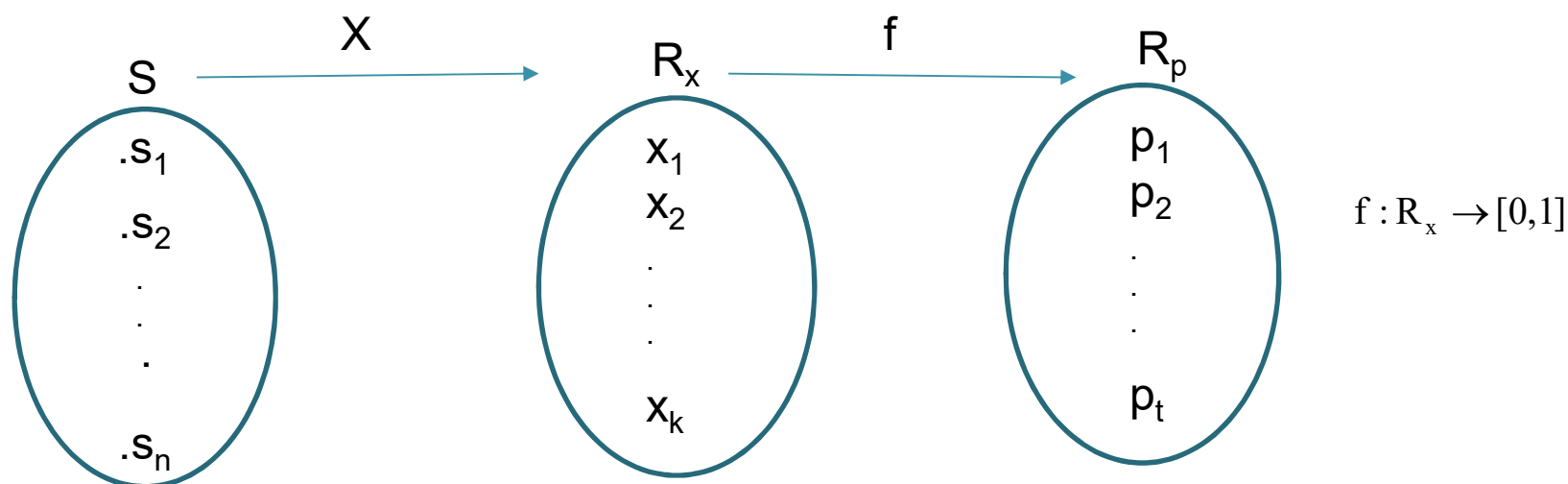
<i>Element of sample space</i>	<i>Probability</i>	<i>x</i>	<i>Element of sample space</i>	<i>Probability</i>	<i>x</i>
HHHH	$\frac{1}{16}$	4	HTTH	$\frac{1}{16}$	2
HHHT	$\frac{1}{16}$	3	THHT	$\frac{1}{16}$	2
HHTH	$\frac{1}{16}$	3	THTH	$\frac{1}{16}$	2
HTHH	$\frac{1}{16}$	3	TTHH	$\frac{1}{16}$	2
THHH	$\frac{1}{16}$	3	HTTT	$\frac{1}{16}$	1
HHTT	$\frac{1}{16}$	2	THTT	$\frac{1}{16}$	1
HTHT	$\frac{1}{16}$	2	TTHT	$\frac{1}{16}$	1
			TTHH	$\frac{1}{16}$	1
			TTTT	$\frac{1}{16}$	0

Thus, we can write $P(X = 3) = \frac{4}{16}$, for example, for the probability of the event that the random variable X will take on the value 3.

PROBABILITY FUNCTION

1)PROBABILTY FUNCTION OF A DISCRETE RANDOM VARIABLE:

Definition: Let $X(S) = R_x = \{x_1, x_2, \dots, x_k\}$ be given



The function $f(x_i) = P(X = x_i)$, $i = 1, 2, \dots, k$ which represents the probability that the random variable X is the value x_i is called probability function or probability ditribution.

DEFINITION **PROBABILITY DISTRIBUTION.** *If X is a discrete random variable, the function given by $f(x) = P(X = x)$ for each x within the range of X is called the probability distribution of X .*

Based on the postulates of probability, we obtain the following theorem.

THEOREM 1. A function can serve as the probability distribution of a discrete random variable X if and only if its values, $f(x)$, satisfy the conditions

1. $f(x) \geq 0$ for each value within its domain;
2. $\sum_x f(x) = 1$, where the summation extends over all the values within its domain.

Ex: A probability function of X random variable is given in table,

x_i	1	2	3	4
$f(x_i)$	1/4	1/5	1/20	1/2

obtain $Y=3X+1$ random variable's probability function.

Solution:

y_i	4	7	10	13
$f(y_i)$	1/4	1/5	1/20	1/2

Ex: The probability function of random variable X is as follows.

$$f(x) = \begin{cases} -\frac{1}{3}(x-2) & x = 0, 1, 2 \\ 0 & \text{Otherwise} \end{cases}$$

What is the probability function of the random variable Y, defined as $Y = 2X$?

Solution: $Y=2X$ or $X=Y/2$

$$f(y) = P(Y = y) = \begin{cases} -\frac{1}{3}\left(\frac{y}{2} - 2\right) & y = 0, 2, 4 \\ 0 & \text{Otherwise} \end{cases}$$

Example:

Find a formula for the probability distribution of the total number of heads obtained in four tosses of a balanced coin.

Solution

Based on the probabilities in the table, we find that $P(X = 0) = \frac{1}{16}$, $P(X = 1) = \frac{4}{16}$, $P(X = 2) = \frac{6}{16}$, $P(X = 3) = \frac{4}{16}$, and $P(X = 4) = \frac{1}{16}$. Observing that the numerators of these five fractions, 1, 4, 6, 4, and 1, are the binomial coefficients $\binom{4}{0}$, $\binom{4}{1}$, $\binom{4}{2}$, $\binom{4}{3}$, and $\binom{4}{4}$, we find that the formula for the probability distribution can be written as

$$f(x) = \frac{\binom{4}{x}}{16} \quad \text{for } x = 0, 1, 2, 3, 4$$

Example:

Check whether the function given by

$$f(x) = \frac{x+2}{25} \quad \text{for } x = 1, 2, 3, 4, 5$$

can serve as the probability distribution of a discrete random variable.

Solution

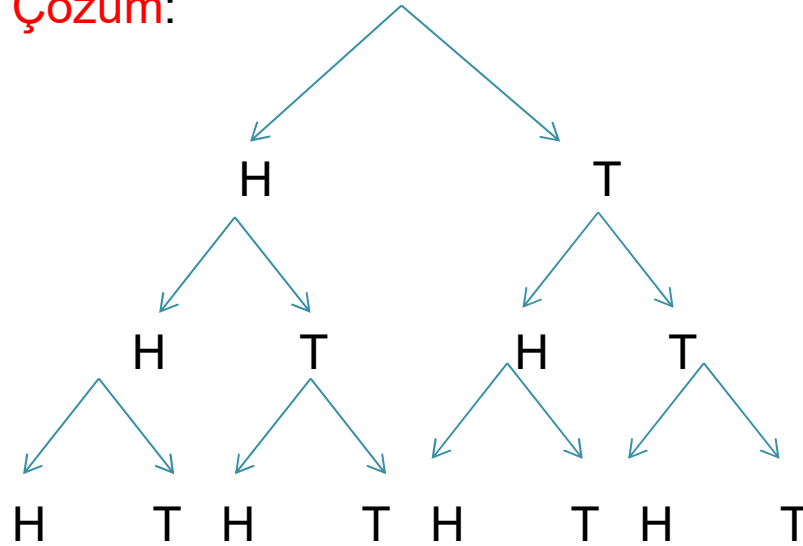
Substituting the different values of x , we get $f(1) = \frac{3}{25}$, $f(2) = \frac{4}{25}$, $f(3) = \frac{5}{25}$, $f(4) = \frac{6}{25}$, and $f(5) = \frac{7}{25}$. Since these values are all nonnegative, the first condition of Theorem 1 is satisfied, and since

$$\begin{aligned} f(1) + f(2) + f(3) + f(4) + f(5) &= \frac{3}{25} + \frac{4}{25} + \frac{5}{25} + \frac{6}{25} + \frac{7}{25} \\ &= 1 \end{aligned}$$

the second condition of Theorem 1 is satisfied. Thus, the given function can serve as the probability distribution of a random variable having the range $\{1, 2, 3, 4, 5\}$.

Ex: 3 coins are tossed together. If the number of heads are represented by the random variable X , determine the probability function.

Çözüm:



$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

$R_x = \{0, 1, 2, 3\}$

X	0	1	2	3
$f(x)$	1/8	3/8	3/8	1/8

$$f(x) = \begin{cases} 1/8, & x = 0 \\ 3/8, & x = 1 \\ 3/8, & x = 2 \\ 1/8, & x = 3 \end{cases}$$

Ex: Determine that $f(x) = \frac{2x}{k(k+1)}$, $x = 1, 2, \dots, k$ is a probability function?

Solution:

Ex: Let $f(x) = \begin{cases} 2kx, & x = 1, 2, 3 \\ k(3+x), & x = 4, 5 \\ 0, & \text{otherwise} \end{cases}$

- a) Find the value of k if f is a probability function,
- b) Compute $P(X=8)$, $P(X=3)$ and $P(x \geq 2 | x \leq 4)$

Ex: If X is defined as the number of boys in a family with 3 children, find the probability function of X . Calculate the probability that the family has at least one boy.

2) PROBABILITY FUNCTION OF A CONTINUOUS RANDOM VARIABLE

Definition: The function f is called the **probability density function** of the continuous random variable X , if

1) $f(x) \geq 0$ for all x values

2) $\int_{-\infty}^{\infty} f(x)dx = 1$

are satisfied.

The probability $P(a \leq X \leq b) = \int_a^b f(x) dx$

For a constant a

$P(X = a) = P(a \leq X \leq a) = \int_a^a f(x) dx = 0$  **It means that the probability of a single point is zero for continuous variables**

Thus,

$$P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b)$$

Example: If $f(x) = \begin{cases} kx^2, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$ is a probability function, find the value of k .

Solution:

Example:

If X has the probability density

$$f(x) = \begin{cases} k \cdot e^{-3x} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

find k and $P(0.5 \leq X \leq 1)$.

Solution:

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^{\infty} k \cdot e^{-3x} dx = k \cdot \lim_{t \rightarrow \infty} \frac{e^{-3x}}{-3} \Big|_0^t = \frac{k}{3} = 1$$

and it follows that $k = 3$. For the probability we get

$$P(0.5 \leq X \leq 1) = \int_{0.5}^1 3e^{-3x} dx = -e^{-3x} \Big|_{0.5}^1 = -e^{-3} + e^{-1.5} = 0.173$$

CUMULATIVE DISTRIBUTION (DISTRIBUTION FUNCTION):

Definition: The cumulative distribution $F(x)$ of a **discrete random** variable X with probability function $f(x)$ is

$$F(x) = P(X \leq x) = \sum_{u \leq x} f(u) \quad \text{for } -\infty < x < \infty$$

If X is a **continuous random** variable, then

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du \quad \text{for } -\infty < x < \infty$$

Ex:

x	0	1	2
f(x)	0.15	0.50	0.35

Determine the cumulative function F and plot it.

Solution:

Some Properties of Cumulative Distribution of a Discrete Variable

- a) $F(x)$; is a non-decreasing step function in the interval $[0,1]$
- b) $P(X > a) = 1 - P(X \leq a) = 1 - F(a)$
- c) $P(a < X \leq b) = F(b) - F(a)$

Some Properties of Cumulative Distribution of a Continuous Variable

- a) $\lim_{x \rightarrow -\infty} F(x) = F(-\infty) = 0$
- b) $\lim_{x \rightarrow \infty} F(x) = F(\infty) = 1$
- c) $F(x)$; is a non-decreasing function in the interval $[0,1]$.
- d) $\frac{d}{dx} F(X) = f(x)$ $(F'(X) = f(x))$
- e) $P(X > a) = 1 - F(a)$ f) $P(a < X \leq b) = F(b) - F(a) = \int_a^b f(x) dx$

Ex:

Find the distribution function of the total number of heads obtained in four tosses of a balanced coin.

Solution

Given $f(0) = \frac{1}{16}$, $f(1) = \frac{4}{16}$, $f(2) = \frac{6}{16}$, $f(3) = \frac{4}{16}$, and $f(4) = \frac{1}{16}$ from Example it follows that

$$F(0) = f(0) = \frac{1}{16}$$

$$F(1) = f(0) + f(1) = \frac{5}{16}$$

$$F(2) = f(0) + f(1) + f(2) = \frac{11}{16}$$

$$F(3) = f(0) + f(1) + f(2) + f(3) = \frac{15}{16}$$

$$F(4) = f(0) + f(1) + f(2) + f(3) + f(4) = 1$$

Hence, the distribution function is given by

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{1}{16} & \text{for } 0 \leq x < 1 \\ \frac{5}{16} & \text{for } 1 \leq x < 2 \\ \frac{11}{16} & \text{for } 2 \leq x < 3 \\ \frac{15}{16} & \text{for } 3 \leq x < 4 \\ 1 & \text{for } x \geq 4 \end{cases}$$

Observe that this distribution function is defined not only for the values taken on by the given random variable, but for all real numbers. For instance, we can write $F(1.7) = \frac{5}{16}$ and $F(100) = 1$, although the probabilities of getting “at most 1.7 heads” or “at most 100 heads” in four tosses of a balanced coin may not be of any real significance.

Ex:

If the distribution function of X is given by

$$F(x) = \begin{cases} 0 & \text{for } x < 2 \\ \frac{1}{36} & \text{for } 2 \leq x < 3 \\ \frac{3}{36} & \text{for } 3 \leq x < 4 \\ \frac{6}{36} & \text{for } 4 \leq x < 5 \\ \frac{10}{36} & \text{for } 5 \leq x < 6 \\ \frac{15}{36} & \text{for } 6 \leq x < 7 \\ \frac{21}{36} & \text{for } 7 \leq x < 8 \\ \frac{26}{36} & \text{for } 8 \leq x < 9 \\ \frac{30}{36} & \text{for } 9 \leq x < 10 \\ \frac{33}{36} & \text{for } 10 \leq x < 11 \\ \frac{35}{36} & \text{for } 11 \leq x < 12 \\ 1 & \text{for } x \geq 12 \end{cases}$$

find the probability distribution of this random variable.

$$f(2) = \frac{1}{36}, f(3) = \frac{3}{36} - \frac{1}{36} = \frac{2}{36}, f(4) = \frac{6}{36} - \frac{3}{36} = \frac{3}{36}, f(5) = \frac{10}{36} - \frac{6}{36} = \frac{4}{36}, \dots, f(12) = 1 - \frac{35}{36} = \frac{1}{36},$$

Example: Let X be a random variable that indicates the charging time of a battery. If the probability density function;

$$f(x) = \begin{cases} \frac{1}{2}e^{-x/2} & x > 0 \\ 0 & \text{Other Cases} \end{cases}$$

- What is the probability that this battery will charge in less than 1 hour or more than 3 hours?
- What is the probability that this time will be longer than 2 hours given that the battery is known to be charged in more than 1 hour?
- What is $F(x)$?

Solution: A: Charging time less than 1 hour
B: Charging time is more than 3 hours

a)

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) \\ &= \int_0^1 \frac{1}{2}e^{-x/2} dx + \int_3^{\infty} \frac{1}{2}e^{-x/2} dx \\ &= 1 - e^{-1/2} + e^{-3/2} \\ &= 1 - 0,61 + 0,22 \\ &= 0,61 \end{aligned}$$

b)

$$b) P(X > 2 \mid X > 1) = \frac{P(X > 2)}{P(X > 1)}$$

$$\frac{P(X > 2)}{P(X > 1)} = \frac{\int_2^{\infty} \frac{1}{2}e^{-x/2} dx}{\int_1^{\infty} \frac{1}{2}e^{-x/2} dx} = \frac{e^{-1}}{e^{-1/2}} = e^{-1/2} = 0,61$$

c)

$$\begin{aligned} \text{c) } F(x) &= P(X \leq x) = \int_0^x \frac{1}{2} \cdot e^{-\frac{x}{2}} dx \\ &= \frac{1}{2} (-2) e^{-x/2} \Big|_0^x \\ &= 1 - e^{-x/2} \end{aligned}$$

$$F(x) = \begin{cases} 1 - e^{-x/2} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

Example: Cumulative function of X is given by

$$F(x) = \begin{cases} 0 & x \leq 0 \\ x(2-x) & 0 < x < 1 \\ 1 & 1 \leq x \end{cases}$$

If $Y=2X+1$, then find the cumulative function of G(y) and probability density function g(y).

Solution:

$$\begin{aligned} G(y) &= P(Y \leq y) = P(2X+1 \leq y) \\ &= P\left(x \leq \frac{y-1}{2}\right) \\ &= F\left(\frac{y-1}{2}\right) \end{aligned}$$

$$G(y) = \begin{cases} 0 & y \leq 1 \\ \frac{y-1}{2} \left(2 - \frac{y-1}{2} \right) = \frac{-y^2 + 6y - 5}{4} & 1 < y < 3 \\ 1 & 3 \leq y \end{cases}$$

$$G'(y) = g(y)$$

$$g(y) = \frac{dG(y)}{dy} = \frac{d\left[(-y^2 + 6y - 5)/4\right]}{dy} = \frac{1}{4}(-2y + 6)$$

$$g(y) = \frac{1}{2}(-y + 3)$$

$$g(y) = \begin{cases} -\frac{y}{2} + \frac{3}{2}, & 1 < y < 3 \\ 0, & \text{elsewhere} \end{cases}$$

Ex: If $F(x) = \begin{cases} 0, & x \leq 0 \\ x, & 0 < x < 1 \\ 1, & x \geq 1 \end{cases}$, find $f(x)$.

Solution:

For $0 < x < 1$ $\frac{dF}{dx} = f(x) \Rightarrow \frac{d(x)}{dx} = 1 = f(x)$

$$f(x) = \begin{cases} 0, & x \leq 0 \\ 1, & 0 < x < 1 \\ 0, & x \geq 1 \end{cases}$$

Example: Probability function of discrete random variable of X is given below;

$X=x:$	-3	0	2	3
$P(X=x):$	0,2	0,1	0,4	c

- Find the value of c?
- At which value, X's probability maximum?
- $P(X>0)=?$
- $P(X=-2)=?$
- Find the cumulative function of X?
- Show F(x) as a Table.
- Draw the Graphs of f(x) and F(x)

Solution:

a) $0,2+0,1+0,4+c=1 \implies c=0,3$

b) 2

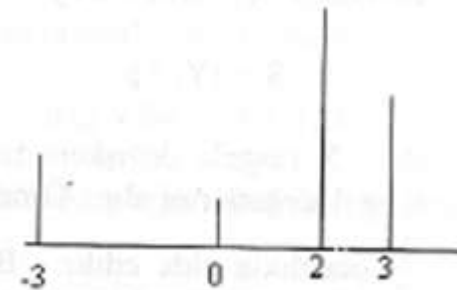
c) $P(X>0) = P(X=2)+P(X=3)=0,4+0,3=0,7.$

d) $P(X = -2) = 0.$

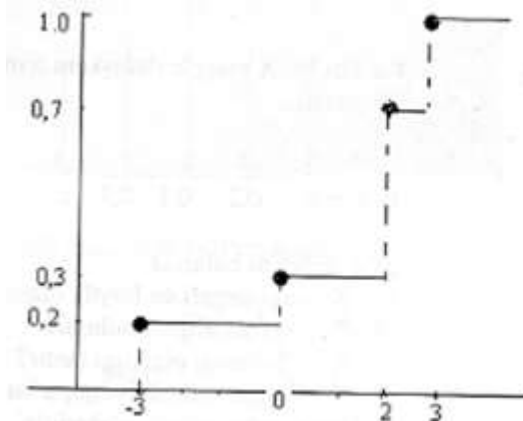
e)
$$F(x) = \begin{cases} 0 & x < -3 \\ 0,2 & -3 \leq x < 0 \\ 0,3 & 0 \leq x < 2 \\ 0,7 & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

f)

$X=x:$	-3	0	2	3
$F(x)=P(X \leq x):$	0,2	0,3	0,7	1.0



Graphs of f(x)



Graphs of cumulative function of F(x)

Example: A coin is tossed until a head occurs. Let X be a random variable that shows necessary number of tossing. Find the probability function of random variable X.

Solution:

		Probability
First toss	H	1/2
Second toss	TH	1/2.1/2
Third toss	TTH	1/2.1/2.1/2
.	.	.
N-th toss	TTT...TH	$(1/2)^{N-1}.1/2$
.	.	.

X = x	1	2	...	x	...
$f(x) = P(X=x)$	$\frac{1}{2}$	$\left(\frac{1}{2}\right)^2$	...	$\left(\frac{1}{2}\right)^x$	...

$$f(x) = \left(\frac{1}{2}\right)^x, \quad x = 1, 2, \dots$$

Note: $a + ar + ar^2 + \dots = \frac{a}{1-r}$

$$(-1 \leq r \leq 1)$$

$$\sum_{x=1}^{\infty} \left(\frac{1}{2}\right)^x = \frac{1}{2} + \frac{1}{2^2} + \dots$$

$$a = \frac{1}{2}, r = \frac{1}{2} \Rightarrow \sum_{x=1}^{\infty} f(x) = \frac{1/2}{1-1/2} = 1$$

EXPECTED VALUE:

Definition: Let X be a random variable with probability function $f(x)$. The expected value (or mean) of X is

$$E(X) = \mu_X = \sum_{i=1}^N x_i f(x_i) \quad \Rightarrow \quad \text{if } X \text{ is discrete}$$

$$E(X) = \mu_X = \int_{-\infty}^{\infty} x \cdot f(x) dx \quad \Rightarrow \quad \text{if } X \text{ is continuous.}$$

Theorem: If $g(x)$ is a function of random variable X with probability function $f(x)$, then expected value of $g(x)$ is

$$E(g(x)) = \mu_{g(x)} = \begin{cases} \sum_{i=1}^n g(x_i) f(x_i), & \text{for discrete variable} \\ \int_{-\infty}^{\infty} g(x) f(x) dx, & \text{for continuous variable} \end{cases}$$

Properties of Expected Value

c is a constant, X and Y are random variables;

1) $E(c)=c$

2) $E(X+c)=E(X)+c$

3) $E(cX)=cE(X)$

4) $x \geq 0 \Rightarrow E(X) \geq 0$

5) $x \leq y \Rightarrow E(X) \leq E(Y)$

6) $E(X+Y) = E(X) + E(Y)$

7) $|E(X)| \leq E(|X|)$

8) Under the assumption that X and Y are independent

$$E(XY) = E(X) E(Y)$$

9) $E(X^2) \neq [E(X)]^2$

10) The expected value E is a linear operator, where a and b constant.

$$E(aX+b) = a E(X) + b$$

Example: Prove that the expected value E is a linear operator, where a and b constant.

$$E(aX + b) = a E(X) + b$$

Solution:

$$E(aX + b) = \sum (aX + b)f(x) = a \sum Xf(x) + b \sum f(x) = aE(x) + b$$

Example: Show $E(cX) = cE(X)$

Solution:

$$\begin{aligned} E(cX) &= \sum_{i=1}^k cx_i P(X = x_i) \\ &= c \sum_{i=1}^k x_i P(X = x_i) \\ &= cE(x) \end{aligned}$$

Example:

X	0	1	2
f(x)	1/2	1/4	1/4

a) $E(X)=?$

b) If $Y=g(X)=2x^2+x$, then find $E(Y)=?$

Solution:

Example: In a game, a coin is successively tossed until it shows a tail or head 3 times. If the game is over with 3 or more trials, then the player will win 15 tl, otherwise the player will lose 5 tl. Find the expected value of X that denotes the gain or loss of the player?

Solution:



Example:

According to a game, a die is rolled and if it shows a prime number, the player wins two times of the number. Otherwise, the player loses 1.9 times of the showing number. Would the player play this game?

Solution:

EXAMPLE

If X is the number of points rolled with a balanced die, find the expected value of $g(X) = 2X^2 + 1$.

Solution

Since each possible outcome has the probability $\frac{1}{6}$, we get

$$\begin{aligned} E[g(X)] &= \sum_{x=1}^6 (2x^2 + 1) \cdot \frac{1}{6} \\ &= (2 \cdot 1^2 + 1) \cdot \frac{1}{6} + \cdots + (2 \cdot 6^2 + 1) \cdot \frac{1}{6} \\ &= \frac{94}{3} \end{aligned}$$

EXAMPLE 4

If X has the probability density

$$f(x) = \begin{cases} e^{-x} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

find the expected value of $g(X) = e^{3X/4}$.

Solution:

$$\begin{aligned} E[e^{3X/4}] &= \int_0^{\infty} e^{3x/4} \cdot e^{-x} dx \\ &= \int_0^{\infty} e^{-x/4} dx \\ &= 4 \end{aligned}$$

Example:

If the probability density of X is given by

$$f(x) = \begin{cases} 2(1-x) & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

(a) show that

$$E(X^r) = \frac{2}{(r+1)(r+2)}$$

(b) and use this result to evaluate

$$E[(2X+1)^2]$$

Solution

(a)

$$\begin{aligned} E(X^r) &= \int_0^1 x^r \cdot 2(1-x) dx = 2 \int_0^1 (x^r - x^{r+1}) dx \\ &= 2 \left(\frac{1}{r+1} - \frac{1}{r+2} \right) = \frac{2}{(r+1)(r+2)} \end{aligned}$$

(b) Since $E[(2X+1)^2] = 4E(X^2) + 4E(X) + 1$ and substitution of $r = 1$ and $r = 2$ into the preceding formula yields $E(X) = \frac{2}{2 \cdot 3} = \frac{1}{3}$ and $E(X^2) = \frac{2}{3 \cdot 4} = \frac{1}{6}$, we get

$$E[(2X+1)^2] = 4 \cdot \frac{1}{6} + 4 \cdot \frac{1}{3} + 1 = 3$$

Example:

Show that

$$E[(aX + b)^n] = \sum_{i=0}^n \binom{n}{i} a^{n-i} b^i E(X^{n-i})$$

Solution

Since $(ax + b)^n = \sum_{i=0}^n \binom{n}{i} (ax)^{n-i} b^i$, it follows that

$$\begin{aligned} E[(aX + b)^n] &= E \left[\sum_{i=0}^n \binom{n}{i} a^{n-i} b^i X^{n-i} \right] \\ &= \sum_{i=0}^n \binom{n}{i} a^{n-i} b^i E(X^{n-i}) \end{aligned}$$

Example:

A lot of 12 television sets includes 2 with white cords. If 3 of the sets are chosen at random for shipment to a hotel, how many sets with white cords can the shipper expect to send to the hotel?

Solution

Since x of the 2 sets with white cords and $3 - x$ of the 10 other sets can be chosen in $\binom{2}{x} \binom{10}{3-x}$ ways, 3 of the 12 sets can be chosen in $\binom{12}{3}$ ways, and these $\binom{12}{3}$ possibilities are presumably equiprobable, we find that the probability distribution of X , the number of sets with white cords shipped to the hotel, is given by

$$f(x) = \frac{\binom{2}{x} \binom{10}{3-x}}{\binom{12}{3}} \quad \text{for } x = 0, 1, 2$$

or, in tabular form,

x	0	1	2
$f(x)$	$\frac{6}{11}$	$\frac{9}{22}$	$\frac{1}{22}$

Now,

$$E(X) = 0 \cdot \frac{6}{11} + 1 \cdot \frac{9}{22} + 2 \cdot \frac{1}{22} = \frac{1}{2}$$

and since half a set cannot possibly be shipped, it should be clear that the term “expect” is not used in its colloquial sense. Indeed, it should be interpreted as an average pertaining to repeated shipments made under the given conditions.